CBSE MOCK — Pre-Board Practice Series

Central Board of Secondary Education

CLASS XII — MATHEMATICS

Date: 12 November 2026 | Maximum Marks: 80 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 5 sections. Read the instructions for each section carefully.
2. All questions are compulsory unless stated otherwise.
3. Write all answers neatly and show relevant working wherever required.

(i) Calculators are NOT permitted.

4. Marks for each question are indicated against it.
5. This paper is a Practice Paper — it is similar to but NOT an actual CBSE board paper.
6. Download more FREE practice papers daily at: **www.cbsemock.org.in**

SECTION A — MCQ & Assertion-Reason (20 Marks)

MCQ (1 mark each) | A/R (1 mark each)

Q1. **[1 mark]**

The value of $|\hat{i} \times \hat{j}| + |\hat{j} \times \hat{k}| + |\hat{k} \times \hat{i}|$ is:

(A) 0
(B) 1
(C) 2
(D) 3

Q2. **[1 mark]**

The function $f(x) = |x - 1| + |x + 1|$ is:

(A) differentiable everywhere
(B) not differentiable at $x = 1$ and $x = -1$
(C) continuous only at $x = 0$
(D) discontinuous at $x = 0$

Q3. **[1 mark]**

The number of all possible matrices of order 3×3 with each entry 0 or 1 is:

(A) 27
(B) 18
(C) 81
(D) 512

Q4. **[1 mark]**

Assertion (A): For two non-zero vectors $\vec{a}$ and $\vec{b}$, $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2 \cdot |\vec{b}|^2$. Reason (R): $\vec{a} \times \vec{b}$ and $\vec{a} \cdot \vec{b}$ are related by $\sin^2\theta + \cos^2\theta = 1$.

Q5. **[1 mark]**

If $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, and $f(0) = 0$, then f is:

(A) continuous but not differentiable at $x = 0$
(B) differentiable at $x = 0$
(C) not continuous at $x = 0$
(D) continuous and differentiable at every point

Q6. **[1 mark]**

If P(A) = 0.4, P(B) = 0.3 and P(A ∩ B) = 0.2, then P(B/A) is:

(A) 1/2
(B) 2/3
(C) 1/3
(D) 3/4

Q7. **[1 mark]**

If f: R → R is defined by f(x) = x^2 + 1, then f^{-1}(10) is:

(A) {3, −3}
(B) {3}
(C) {−3}
(D) ∅

Q8. **[1 mark]**

Assertion (A): The function f(x) = |cos x| is not differentiable at x = π/2. Reason (R): The function |x| is not differentiable at x = 0.

Q9. **[1 mark]**

∫ e^x(sin x + cos x) dx =

(A) e^x sin x + C
(B) e^x cos x + C
(C) e^x(sin x + cos x) + C
(D) e^x(sin x − cos x) + C

Q10. **[1 mark]**

If the matrix A = [[0, a, −3], [2, 0, −1], [b, 1, 0]] is skew-symmetric, then:

(A) a=2, b=3
(B) a=−2, b=−3
(C) a=2, b=−3
(D) a=−2, b=3

Q11. **[1 mark]**

The degree of the differential equation $(d^2y/dx^2)^3$ + $(dy/dx)^2$ + sin(dy/dx) + 1 = 0 is:

(A) 3
(B) 2
(C) 1
(D) not defined

Q12. **[1 mark]**

The shortest distance between the lines $\vec{r}$ = ($\hat{i}$ + 2$\hat{j}$ + 3$\hat{k}$) + λ(2$\hat{i}$ + 3$\hat{j}$ + 4$\hat{k}$) and $\vec{r}$ = (2$\hat{i}$ + 4$\hat{j}$ + 5$\hat{k}$) + μ(3$\hat{i}$ + 4$\hat{j}$ + 5$\hat{k}$) is:

(A) 1/√6
(B) 1/√3
(C) √6
(D) 1/√2

Q13. **[1 mark]**

$\int_0^{\pi/2}$ cos x · $e^{\sin x}$ dx =

(A) e − 1
(B) e + 1
(C) e
(D) 1

Q14. **[1 mark]**

The area of the region bounded by the curve $y = x^2$, the x-axis, and $x = 1$ and $x = 3$ is:

(A) 26/3 sq. units
(B) 8/3 sq. units
(C) 10 sq. units
(D) 28/3 sq. units

Q15. **[1 mark]**

If A is a square matrix such that $A^2 = A$, then $(I + A)^3 - 7A$ is equal to:

(A) A
(B) I+A
(C) I
(D) 3A

Q16. **[1 mark]**

If $y = \sin^{-1}(2x\sqrt{(1-x^2)})$, then dy/dx is:

(A) $2/\sqrt{(1-x^2)}$
(B) $-2/\sqrt{(1-x^2)}$
(C) $1/\sqrt{(1-x^2)}$
(D) $-1/\sqrt{(1-x^2)}$

Q17. **[1 mark]**

The number of arbitrary constants in the general solution of a differential equation of order 3 is:

(A) 1
(B) 2
(C) 3
(D) 0

Q18. **[1 mark]**

The principal value of $\sin^{-1}(-\sqrt{3}/2)$ is:

(A) $\pi/3$
(B) $-\pi/3$
(C) $2\pi/3$
(D) $-2\pi/3$

Q19. **[1 mark]**

If the feasible region is bounded, then the linear programming problem has:

(A) no solution
(B) a unique solution at a corner point
(C) optimal solution at a corner point
(D) optimal solution inside the feasible region

Q20. **[1 mark]**

The vector equation of the line passing through the point (1, 2, −4) and perpendicular to the plane $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k}) = 7$ is:

(A) $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} - 3\hat{j} + \hat{k})$
(B) $\vec{r} = (2\hat{i} - 3\hat{j} + \hat{k}) + \lambda(\hat{i} + 2\hat{j} - 4\hat{k})$
(C) $\vec{r} = (\hat{i} + 2\hat{j} + 4\hat{k}) + \lambda(2\hat{i} - 3\hat{j} + \hat{k})$
(D) $\vec{r} = (\hat{i} - 2\hat{j} + 4\hat{k}) + \lambda(2\hat{i} - 3\hat{j} + \hat{k})$

SECTION B — Very Short Answer (10 Marks)

VSA (2 marks each)

Q21. **[2 marks]**

The sides of an equilateral triangle are increasing at the rate of 2 cm/s. Find the rate at which the area is increasing when the side is 10 cm.

Q22. **[2 marks]**

Find the intervals in which the function $f(x) = 2x^3 - 9x^2 + 12x - 5$ is: (a) increasing (b) decreasing

Q23. **[2 marks]**

Find the value of $\sin^{-1}(\sin 3\pi/5)$. OR Prove that: $\tan^{-1}(1/2) + \tan^{-1}(1/5) + \tan^{-1}(1/8) = \pi/4$.

Q24. **[2 marks]**

Find $\int x/(x+1)\,dx$. OR Evaluate: $\int_1^2 (x^2 + 1)\,dx$.

Q25. **[2 marks]**

Find the angle between vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. OR Find the projection of vector $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.

SECTION C — Short Answer (18 Marks)

SA (3 marks each)

Q26. **[3 marks]**

Find the particular solution of the differential equation: $dy/dx + y \cot x = 4x \operatorname{cosec} x$ $(x \neq 0)$, given that $y = 0$ when $x = \pi/2$.

Q27. **[3 marks]**

A manufacturer makes two types of products A and B. Each product requires 3 hours on machine M_1 and 2 hours on machine M_2. Product B requires 1 hour on M_1 and 1 hour on M_2. The machine M_1 is available for 300 hours and M_2 for 240 hours. If the profit from A is ₹60 and from B is ₹40 per unit, find the maximum profit using LPP (formulate only, do not solve graphically).

Q28. **[3 marks]**

Using integration, find the area of the region bounded by $x^2 = 4y$, y-axis, and $y = 1$ and $y = 4$. OR Find the area bounded by the curve $y = x^2 - 4$ and the line $y = 2 - x$.

Q29. **[3 marks]**

A and B throw a die alternately. The one who gets a 6 first wins. Find the probability of B winning if A starts. OR An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted and it is replaced. If 6 balls are drawn with replacement, find the probability that at most 2 are red.

Q30. **[3 marks]**

Solve the system of equations using matrices: $2x - y + z = -2$ $3x + 2y - 2z = 2$ $x - 2y + z = 3$

Q31. **[3 marks]**

Find $\int (x^2 + x + 1)/((x+2)(x^2 + 1))\,dx$ using partial fractions.

SECTION D — Long Answer (20 Marks)

LA (5 marks each)

Q32. **[5 marks]**

(a) Show that the relation R on the set A = {1,2,3,4,5} given by R = {(a,b): |a−b| is divisible by 2} is an equivalence relation. (b) Find the equivalence classes of elements 1 and 2.

Q33. **[5 marks]**

Find the equation of the plane passing through the line of intersection of planes x + y + z = 1 and 2x + 3y + 4z = 5 and perpendicular to the plane x − y + z = 0. Also find the distance of this plane from the origin. OR Find the foot of perpendicular from P(1,2,3) to the line (x−6)/3 = (y−7)/2 = (z−7)/−2. Also find the length of the perpendicular.

Q34. **[5 marks]**

If $y = (\tan^{-1} x)^2$, show that $(1+x^2)^2 \, d^2y/dx^2 + 2x(1+x^2) \, dy/dx = 2$. OR If $x = a(\theta - \sin\theta)$ and $y = a(1 - \cos\theta)$, find d^2y/dx^2 at $\theta = \pi/2$.

Q35. **[5 marks]**

Using the method of integration, find the area of the triangle whose vertices are (1,0), (2,2) and (3,1). OR Using integration, find the area of the region: $\{(x,y): 0 \le y \le x^2, 0 \le y \le x + 2, 0 \le x \le 3\}$.

SECTION E — Case Study (12 Marks)

CBQ (4 marks each — Case Study)

Q36. **[4 marks]**

A manufacturer produces two types of lamps — regular (R) and deluxe (D). The profit per unit is ₹5 for R and ₹8 for D. The constraints are: labour (R + 2D ≤ 100 hrs), machine time (3R + D ≤ 90 hrs), and both R, D ≥ 0. (a) Write the objective function. (b) Find the corner points of the feasible region. (c) Find the maximum profit. OR (c) If the profit per regular lamp increases to ₹7, does the optimal solution change?

Q37. **[4 marks]**

In a bag there are 5 red, 3 blue, and 2 green balls. Two balls are drawn successively without replacement. (a) Find the probability that both balls are red. (b) Find the probability that the first ball is red and second is blue. (c) Find the probability that both balls are of different colours. OR (c) Find the probability that at least one ball is red.

Q38. **[4 marks]**

In a school project, students studied the spread of a disease. The fraction of people infected at time t (days) follows: P(t) = 1 − e^(−0.1t). (a) Find the rate of change of infection at t = 10 days. (b) At what time is the infection rate maximum? (c) Find the fraction infected at t = 20 days. (Use $e^2 \approx 7.389$)

✦ END OF QUESTION PAPER ✦

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